MIT 8103 Advanced Database Systems

Solar Mini-Grid Energy Distribution and Meter Monitoring System · Aisha Abubakar · Student ID 301924398

Case study

My case study is Solar Mini-Grid Energy Distribution and Meter Monitoring System. I designed the database for mini-grid operators and monitoring analysts, with separate tables named mini_grid, meter and energy_reading. It supports these operations: register a meter to a mini-grid using its unique serial number; record timestamped energy readings and review meter status; inspect readings by grid and time range, and summarize energy by grid and day.

Portfolio 1: Database Design and Modelling

I organised the relational design into 3 tables. Primary keys identify records, foreign keys preserve the links, and checks enforce the stated business rules. The cumulative kWh reading must be zero or greater. The normalisation examples show the sample data at 1NF, 2NF and 3NF. All 9 deliberately invalid inserts were rejected without changing the valid sample rows.

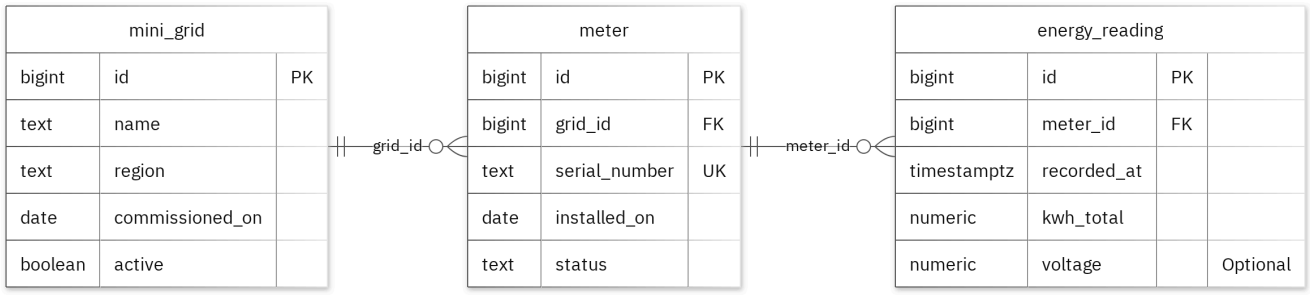


Figure 1. Tables, keys and relationships. PK: primary key; FK: foreign key; UK: unique value.

Portfolio 2: Query Processing and Optimisation

I ran four SQL queries against the sample records. The join returned 2 rows and included Sunrise Mini-Grid with MTR-SUN-001. For the index comparison, I used 20,000 separate performance rows. The median lookup time changed from 1.003 ms to 0.191 ms. Both phases returned the same records. The measured plan used the new index. These timings apply to this lookup and environment.

Portfolio 3: Transactions and Concurrency

I used a temporary counter on meter to test COMMIT, ROLLBACK and READ COMMITTED isolation. The uncommitted update remained hidden from the other session, and ROLLBACK restored the earlier committed value. Without locking, two increments left the counter at 2. With SELECT FOR UPDATE, Session B waited for Session A and the final value was 3. The helper is confined to this exercise; the case's original business fields stay unchanged.

Portfolio 4: NoSQL and Advanced Data Models

I used meter_events for the case's event documents. MongoDB inserted 4 sample events and returned the expected results for 4 queries. Validation rejected an invalid date type and a duplicate event ID. Embedded snapshots keep related details with an event, while the details field can differ by event type. These are fictional events; PostgreSQL continues to enforce its relational rules.

Repository and evidence

https://github.com/aaaesher/Abubakar_Aisha_301924398

The portfolio folders contain the SQL and JavaScript, explanations, query plans, result tables and screenshots.

Portfolio 5: Distributed and Cloud Database Exercise

I proposed three MongoDB replica members across separate failure zones. The discussion covers primary elections, majority acknowledgement, secondary lag and sharding trade-offs for meter_events. Replication supports availability, while backups address recovery. PostgreSQL needs its own availability plan. This is the permitted discussion option; no live replica set or failover test was deployed.

Individual reflection

I kept mini_grid, meter, energy_reading in separate tables so their details could be maintained without repeating them in every related record. The cumulative kWh reading must be zero or greater. The constraint tests check this alongside the required values and record links. One issue illustrated by the two-session exercise is that both sessions can read the same value before either update is committed. Locking the row before reading preserves both increments. The index comparison also shows why I need to check the execution plan as well as the final answer: the same result can require different amounts of database work.

AI-use declaration

I used GitHub Copilot and the portfolio generator to help prepare the database model, scripts and explanations. The generator ran the PostgreSQL and MongoDB exercises, checked their outputs and captured the result images. I remain responsible for checking and explaining the submitted work.

References

- PostgreSQL: constraints
- PostgreSQL: EXPLAIN
- PostgreSQL: isolation
- PostgreSQL: row locks
- MongoDB: data modelling
- MongoDB: replication
- MongoDB: sharding
- Mermaid: ER diagrams